

KAIRON: AI-Powered Career Digital Twin

Domain

Artificial Intelligence | Generative AI | Career Guidance

Problem Description

Students and professionals gain skills through projects, certifications, internships, coding practice, and work experience. However, this information is usually spread across resumes, GitHub profiles, certificates, and other platforms.

Most career guidance systems provide general suggestions and do not understand a person's actual background, skills, and achievements.

There is a need for an intelligent system that can understand a user's complete professional journey and provide personalized career guidance.

Objective

The objective of this project is to build an AI-powered Career Digital Twin that helps users understand their strengths, identify skill gaps, prepare for interviews, and plan their career growth.

The system aims to:

- Build a digital profile of the user
 - Store skills, projects, certifications, and experiences
 - Provide personalized career advice
 - Identify missing skills for target job roles
 - Generate interview questions
 - Recommend learning resources and projects
-

Proposed Solution

The proposed system collects information from:

- Resume
- Projects
- Certifications
- GitHub Profile

- Job Descriptions

The data is stored in a knowledge base and processed using Retrieval-Augmented Generation (RAG).

When users ask questions such as:

- "Am I ready for an AI Engineer role?"
- "What skills am I missing?"
- "Generate interview questions for me."

The system retrieves relevant information from the user's profile and generates personalized answers.

System Workflow

1. Data Collection

Collect information from resumes, projects, certifications, and GitHub.

2. Data Processing

Extract skills, technologies, experiences, and achievements.

3. Knowledge Storage

Store information in a vector database and knowledge graph.

4. Information Retrieval

Retrieve relevant information based on user queries.

5. AI Response Generation

Generate career guidance and recommendations using an LLM.

6. Career Dashboard

Display skills, growth progress, and readiness scores.

Expected Outcome

The system will:

- Provide personalized career guidance
- Identify skill gaps
- Recommend learning paths
- Generate interview preparation questions

- Track career growth over time
 - Help users make better career decisions
-

Innovation & Impact

- Creates a digital twin of a user's career journey
 - Uses RAG for personalized responses
 - Provides career guidance based on actual achievements
 - Continuously updates as users gain new skills and experiences
 - Acts as a personal AI career mentor
-

Technology Stack

Frontend - React.js

Backend - FastAPI

Vector Database - Qdrant

Knowledge Graph - Neo4j

LLM - Llama 3

Embeddings - BGE-M3

Conclusion

KAIRON is an AI-powered Career Digital Twin that helps users understand their skills, prepare for interviews, identify career opportunities, and plan their professional growth. By using RAG and Knowledge Graphs, the system provides personalized guidance based on the user's real experiences and achievements.